Executive Summary

This report outlines a targeted marketing strategy to drive Stripe Subscription adoption. Instead of a broad, untargeted email blast, I analyzed product usage combinations and transaction volumes to find our highest-probability "warm leads."

The Target: 1853 unique merchants spanning Business Services, Education, Merchandise, Software, Personal Services and other businesses of various sizes.

The 'Why': These merchants already trust Stripe Checkout have a high propensity to actually convert since ~ 50% of their direct competitors have already adopted Subscriptions.

The Impact: This targeted strategy projects an estimated $99.4 million in influenced recurring volume. This represents a $35 million volume uplift compared to an untargeted baseline campaign.

The Strategy: By sending a localized campaign to these specific users highlighting that they are in the minority of their peer group, we leverage "social proof" to maximize conversion rates, while reserving massive Enterprise clients for direct B2B sales.

Data Preparation

Standardization & Cleaning: I stripped trailing/leading whitespace from categorical text, standardized business sizes to lowercase, and forced country codes to uppercase. Financial volume columns were cast strictly to numeric formats to prevent calculation errors, and exact duplicate rows were dropped.

Additionally, duplicate merchants were also cleaned by keeping their first instances once and removing the rest.

Duplicates of merchants

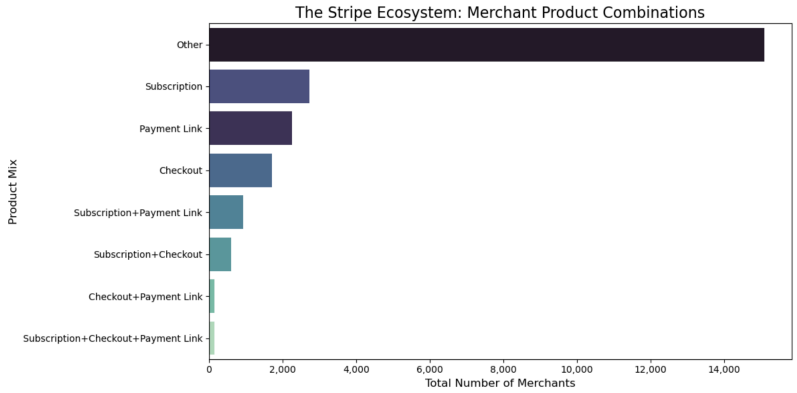
	merchant	industry	first_charge_date	country	business_size
11282	0.00E+00	Healthcare	2040-12-08 00:00:00+00:00	US	small
11606	0.00E+00	Merchandise	2038-11-30 00:00:00+00:00	CA	small
13020	0.00E+00	Education	2041-07-15 00:00:00+00:00	MX	small
13384	0.00E+00	Others	2041-07-24 00:00:00+00:00	GB	small
18458	0.00E+00	Business services	2041-12-16 00:00:00+00:00	US	small
20739	0.00E+00	Others	2042-03-10 00:00:00+00:00	US	small
21151	0.00E+00	Others	2042-03-26 00:00:00+00:00	GB	small
22686	0.00E+00	Leisure	2042-05-22 00:00:00+00:00	SG	small
12221	4.72E+10	Clothing & accessory	2037-08-01 00:00:00+00:00	US	small
13304	4.72E+10	Education	2041-01-20 00:00:00+00:00	IT	small

(10, 5)

Aggregation: The payments dataset contained multiple transactional rows per merchant. To create a cohesive merchant-level view, I aggregated the payments data by grouping by merchant ID. I calculated the sum of total volume for each specific product category and the total transaction count per merchant. Finally, I left-joined the aggregated payment metrics with the merchant dimensional data (industry, business size) to create a clean, unified dataset ready for feature engineering.

I initially considered removing extreme transaction volume outliers to normalize the dataset. However, after doing some preliminary research in fintech, these outliers represent real, highly profitable enterprise clients; rather than deleting them, I chose to isolate them into a separate 'Whale' cohort so they weren't lost from the business's overall view.

Approach and Rationale:
Rather than looking at isolated transactions, I focused the methodology on understanding the merchant's complete product ecosystem. My core strategy was to find "warm leads": merchants who already trust Stripe's ecosystem but haven't adopted the Subscriptions API yet.



Feature Engineering (Product Mix):
I engineered boolean flags (0 or 1) to indicate if a merchant used a specific product, which allowed me to create a composite product_mix category (e.g., "Standalone Checkout", "Subscription + Payment Link").

```
product_mix
Other                                  15087
Subscription                            2729
Payment Link                            2252
Checkout                                1713
Subscription+Payment Link                930
Subscription+Checkout                    600
Checkout+Payment Link                    156
Subscription+Checkout+Payment Link       152
Name: count, dtype: int64
```

Conversion Funnel Analysis: I calculated cross-selling "conversion rates" by isolating merchants who already use baseline products (like Checkout or Payment Links) to see what percentage of them successfully expanded into Subscriptions. Segmenting this by industry and business size highlights natural pathways for product growth.

Targeting Small Businesses (The 2041/2042 Assumption): While Subscriptions is an API-driven product that traditionally required dedicated developer resources and might be difficult for smaller businesses, this dataset reflects future market conditions (2041-2042). I operated under the strategic assumption that the barrier to entry for API implementation will have significantly decreased by this time.

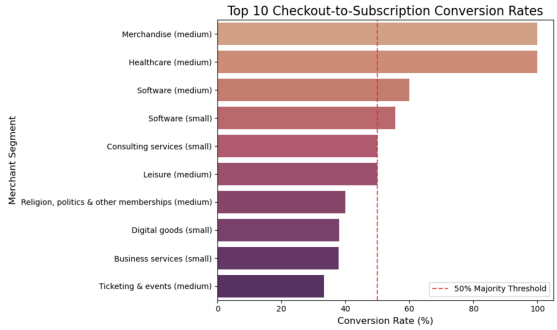
Highest Subscription Adoption Rates for Payment links

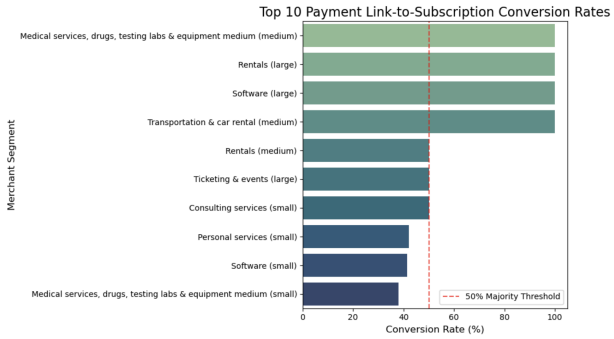
industry	business_size	product_mix	Payment Link	Subscription+Payment Link	Conversion_Rate_%
Medical services, drugs, testing labs & equipment medium	medium		0	1	100.0
Rentals	large		0	1	100.0
Software	large		0	2	100.0
Transportation & car rental	medium		0	1	100.0
Rentals	medium		1	1	50.0
Ticketing & events	large		1	1	50.0
Consulting services	small		6	6	50.0
Personal services	small		249	182	42.2
Software	small		166	118	41.5
Medical services, drugs, testing labs & equipment medium	small		59	36	37.9

Highest Subscription Adoption Rates for Checkout

industry	business_size	product_mix	Checkout	Subscription+Checkout	Conversion_Rate_%
Merchandise	medium		0	2	100.0
Healthcare	medium		0	1	100.0
Software	medium		4	6	60.0
	small		133	166	55.5
Consulting services	small		1	1	50.0
Leisure	medium		1	1	50.0
Religion, politics & other memberships	medium		3	2	40.0
Digital goods	small		91	56	38.1
Business services	small		156	95	37.8
Ticketing & events	medium		2	1	33.3

High-Value Filtering: I isolated the top merchants by their “Conversion_Rate_%” and their non 0 value in either Checkout or Payment Link column. The rationale here is simple: optimizing product adoption among our highest-earning users doesn’t guarantee yielding the highest immediate ROI if they have a very slim chance of converting.





While I initially considered solely using a machine learning algorithm to individually rank every merchant's likelihood to upgrade, that approach is difficult to translate into marketing copy because of the black-box effect. Instead, I grouped merchants by industry and behaviour. This gives the marketing team a clear, persuasive angle for their emails.

Results & Expected Business Impact

Based on the data, I am recommending a targeted campaign aimed primarily at the **Software, Business Services, Education and Merchandise** industries who currently only use Checkout or Payment Links. Full list in a separate .csv file.

Breakdown by Industry

industry	
Business services	560
Others	326
Software	311
Merchandise	277
Personal services	252
Education	127

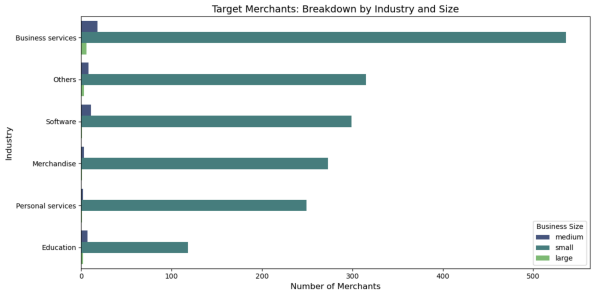
Matrix: Industry vs. Business Size

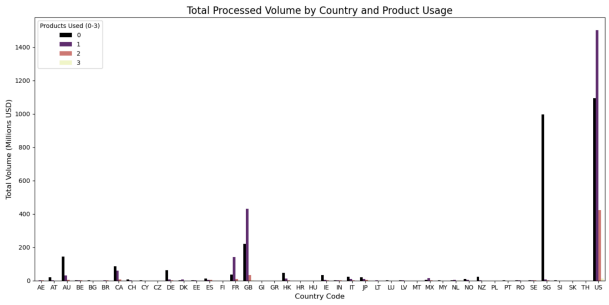
business_size industry	large	medium	small	Total
Business services	6	18	536	560
Education	2	7	118	127
Merchandise	1	3	273	277
Others	3	8	315	326
Personal services	1	2	249	252
Software	1	11	299	311
Total	14	49	1790	1853

- **The Financial Opportunity:** These 1,853 specific merchants roughly process a combined $199 million in total volume.
- **The Targeted Return:** Because ~50% of their direct peers already use Subscriptions, we project a high conversion rate for this campaign. This means we have the potential to influence **$99.4 million** in recurring volume.
- **The Opportunity Cost:** If we ignored the data and sent an untargeted email blast to average small businesses, the baseline conversion rate drops to 35%, yielding only about $64 million in influenced volume.

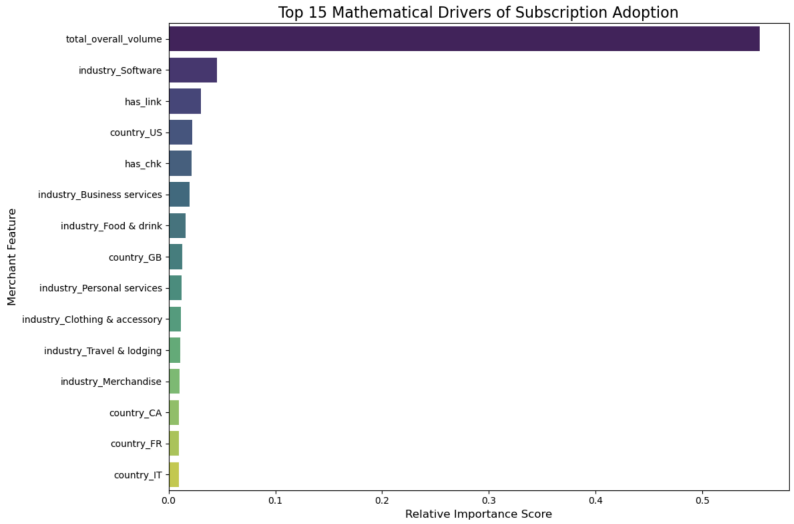
- **The Bottom Line:** By using a targeted, data-driven approach, this campaign generates a potential volume uplift of **$34.95 million** compared to an untargeted baseline campaign.

I could have selected the merchants who are bigger in size. But statistically the most likely to say 'yes' small merchants results in a higher ROI. Geographically, I recommend launching this campaign in the US, Singapore and Great Britain, where the vast majority resides.





Predictive Validation (Machine Learning)



To mathematically validate my manual targeting strategy, I trained a Random Forest model on a sample of merchants to predict subscription adoption. Although the model performed poorly; it confirmed my logic: after total transaction volume (which filters out the enterprise whales), a merchant's industry and existing product mix are the strongest independent predictors of converting to Subscriptions.

Next Steps

If given more resources and time, I would take these specific actions:

- **A/B Test the Campaign:** I would split our target list and test different messaging. One email could highlight the "ease of technical setup" and the other could highlight "recurring revenue stability" to see which drives a higher rate of API integration.
- **Investigate the "0-Product" Whales:** I would request data to see what products those massive retail merchants are actually using (e.g., Stripe Terminal, Invoicing) to build a separate, high-touch enterprise strategy for them.

- **Time Sensitivity:** I would like to explore potential patterns of influences based on the date and time of transactions for each industry. Maybe most subscription volume comes from a specific time and location?
- **Total Volume x Product Mix:** Finally, identifying the total volume earned difference between the combination of products would be interesting to test our assumptions.

Resource & AI Usage

In the interest of transparency and accountability, I utilized Large Language Models (LLMs) during this assessment, specifically the Gemini student account. AI was used strictly as a pair-programming assistant to debug pandas syntax, optimize Matplotlib/Seaborn formatting, and structure the flow of this final document.

All core business logic, feature engineering decisions, targeting strategies, and metric definitions were independently reasoned and derived.

Additionally, I noticed that AI, in general, was focused on pleasing the Head of Product in that scenario/setting rather than understanding and putting Stripe's client at the forefront of decision-making. Finally, if I may be casual, AI can be so useless to understand underlying patterns and explore with curiosity. This assignment was perfectly challenging.

Estimated Time Spent:

2hrs on understanding the provided data, Stripe's motivation and plans
4hrs on executing EDA, visualizations, brainstorming, and rationalizing options
2hrs report writing and executing a quick sample machine learning model

Total: 8hrs